

BACHELOR THESIS
ARTIFICIAL INTELLIGENCE

Radboud University



**Spiking Neurons with Slow Synapses in
Fast Dynamical Systems**

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Abstract

Spiking neural networks are increasingly studied as biologically inspired models for temporal computation and dynamical system control. Sparse spikes enable efficient encoding of low-activity dynamics, while population coding averages noise and distributes uncertainty, leading to robust representations of temporal signals. However, when modeling fast dynamical systems with slow temporal filters, this introduces delays that degrade tracking performance and can destabilize closed-loop feedback systems.

A key development in spike-based modeling of dynamical systems is the framework introduced by Boerlin, Machens, and Denève, in which neural populations represent continuous variables through discrete spike events that act as instantaneous error corrections, and spikes are only emitted when they reduce the error between a target signal and its neural representation. However, this framework is primarily aimed at linear dynamical systems and relies on assumptions about synaptic interactions and temporal processing that limit its applicability to more complex neural computations.

Later work by Nardin et al. introduces structured synaptic interactions, such as multiplicative connections between neural activities. This allows the framework to be extended to nonlinear dynamical systems while preserving the underlying principle of spike-based error correction. As a result, the same event-based coding mechanism can be used to represent a broader class of dynamical processes, significantly extending the computational capabilities of spike-based neural models.

This thesis investigates how synaptic time constants influence the dynamics of spiking neural networks in fast dynamical systems. In particular, it studies the trade-off between biologically plausible slow synaptic filtering and the ability of spike-based systems to accurately track rapidly changing signals. To analyze this, a computational framework is used in which dynamical systems are simulated under varying synaptic time constants, allowing systematic evaluation of how temporal filtering affects tracking accuracy, responsiveness, and stability.

STILL NEED TO ADD RESULTS

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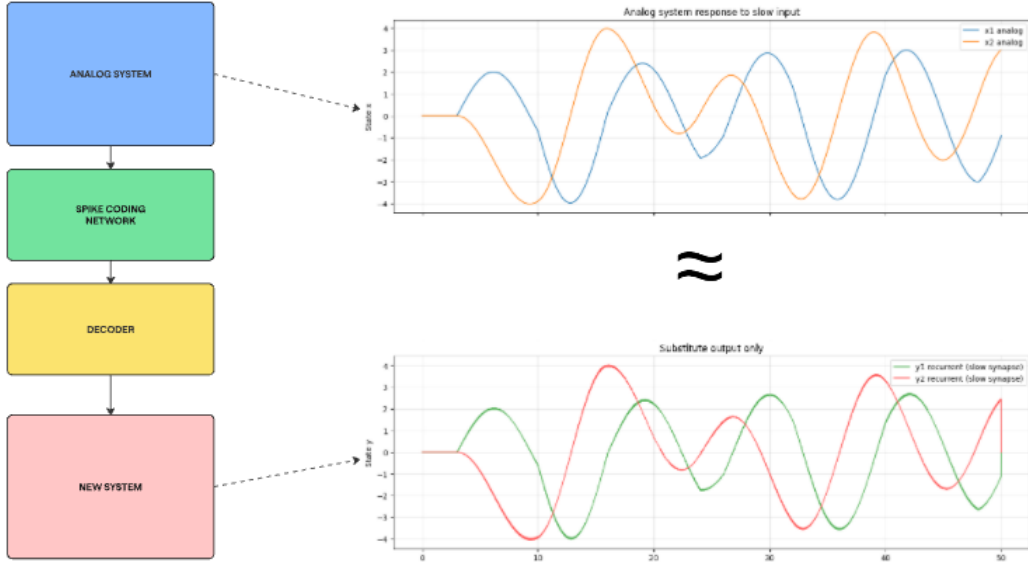
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1 Introduction

Spiking Neural Networks (SNNs) are biologically inspired neural networks that provide an energy-efficient framework for processing temporal information (1). Unlike conventional neural networks, SNNs integrate information over time through spike timing and neural membrane dynamics.

Many real-world processes can be described as dynamical systems, whose states evolve continuously over time. These systems often require fast, accurate prediction and tracking of rapidly changing signals, making temporal processing an interesting component. In particular, closed-loop dynamical systems are of interest because the system’s output directly influences its future inputs. While the system may initially follow the desired dynamics, small errors can accumulate over time, causing deviations from the target trajectory. And thus, accurate tracking and error correction are essential for maintaining stable and reliable performance.

However, when modeling fast dynamical systems with SNNs, a challenge arises. Temporal information in SNNs is often integrated through synaptic filters, but existing artificial synapse models, such as AMPA and NMDA synapses, do not entirely balance responsiveness and memory. The resulting spiking behavior still differs substantially from its biological counterpart, and simply adjusting synaptic time constants is insufficient to resolve this discrepancy. Small synaptic time constants provide rapid responses to changing inputs and enable accurate tracking of fast dynamics, but retain little information over time. Larger synaptic time constants improve temporal integration and produce smoother representations, at the cost of increased delay and reduced responsiveness.



Taken together, these observations highlight a gap in our understanding of how biologically realistic synaptic dynamics interact with the demands of fast dynamical systems. While SNNs are promising options for temporal computation and slow synaptic dynamics provide biologically meaningful temporal integration, fast dynamical systems require rapid and stable tracking to maintain performance. The extent to which slow synaptic filtering limits tracking accuracy and system stability remains insufficiently understood. In this thesis, we study this question using a spike-based coding network based on the framework of Boerlin et al. A population of spiking neurons is used to represent the state of a low-dimensional linear dynamical system. The network decodes the system state from filtered spike trains, allowing direct comparison between the target dynamics and the neural reconstruction. Within this setup, the synaptic time constant is studied to examine its effect on tracking accuracy, delay, spike statistics, and stability. In addition, both open-loop and closed-loop configurations are considered, and a modified spike rule is introduced to account for the delays introduced by slow synaptic filtering.

2 Methods

This section describes the model and methods used to study how spiking neural networks can approximate dynamical systems. We first define the underlying dynamical system that serves as the target signal, after which we introduce the spiking coding framework used to represent and reproduce its state.

2.1 Dynamical System

The initial system is a linear dynamical system of the form

$$\dot{\mathbf{x}} = A\mathbf{x} + B\mathbf{u}, \quad (1)$$

which serves as the baseline analog model before introducing spiking components. This system provides a simple and interpretable framework for studying temporal dynamics while still capturing complex behavior. A low-dimensional state space is used to ensure interpretability and to isolate the effects of synaptic filtering and spiking dynamics without

introducing unnecessary complexity. This allows for a comparison between the reference analog system and its neural representation.

Before defining additional neural components, we first implement a baseline analog simulator. To make the continuous-time system in Eq. (1) compatible with discrete-time simulation and coupling to the spiking network, the dynamics are integrated using a Forward Euler scheme:

$$x(t + \Delta t) = x(t) + \Delta t(Ax(t) + Bu(t)), \quad (2)$$

This discretisation provides a simple and computationally efficient way to approximate the continuous dynamics while enabling step-wise interaction with the spiking network, which operates in the same discrete-time simulation loop.

The system matrix A is chosen to generate damped oscillatory dynamics and is defined as

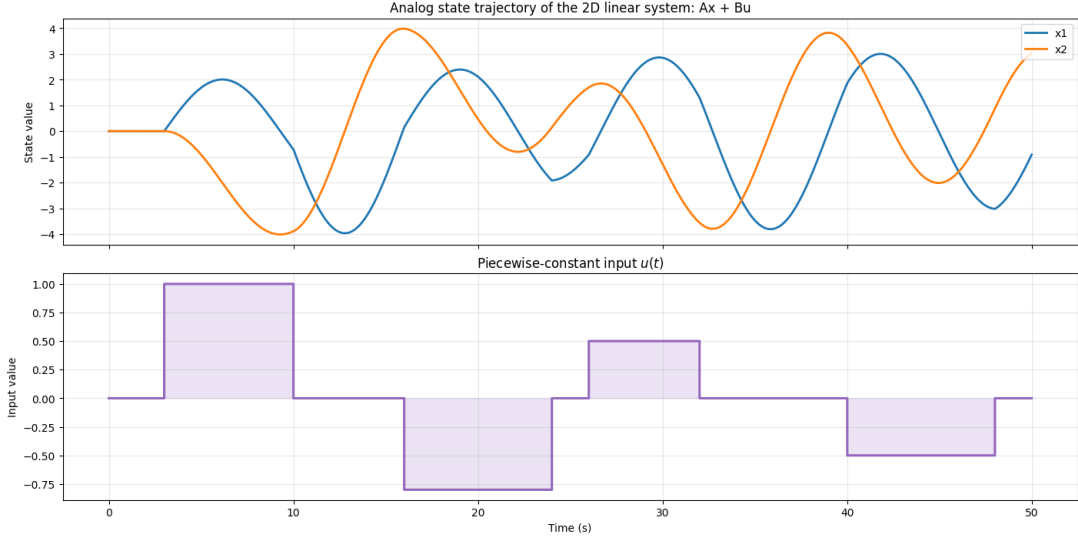
$$A = \begin{pmatrix} -\lambda & \omega \\ -\omega & -\lambda \end{pmatrix}, \quad (3)$$

where $\lambda > 0$ controls the rate of exponential decay (damping), and ω determines the rotational frequency of the system.

The input matrix B is given by

$$B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \quad (4)$$

Both open-loop and closed-loop configurations are considered. In the open-loop setting, the system evolves independently of the neural representation. In contrast, in the closed-loop setting, the decoded output is fed back into the system (see Eq. (1)).



2.2 Spiking coding networks

The goal of the spiking coding network is to represent the continuous state vector $\mathbf{x}$ of the dynamical system introduced in Eq. (1) using only discrete spike events. Rather than storing the state directly, information is distributed across a population of neurons whose collective activity provides a continuous estimate of the system state.

The network follows the spike-based coding framework proposed by Boerlin et al., in which neurons emit spikes whenever doing so reduces the error between the true state and

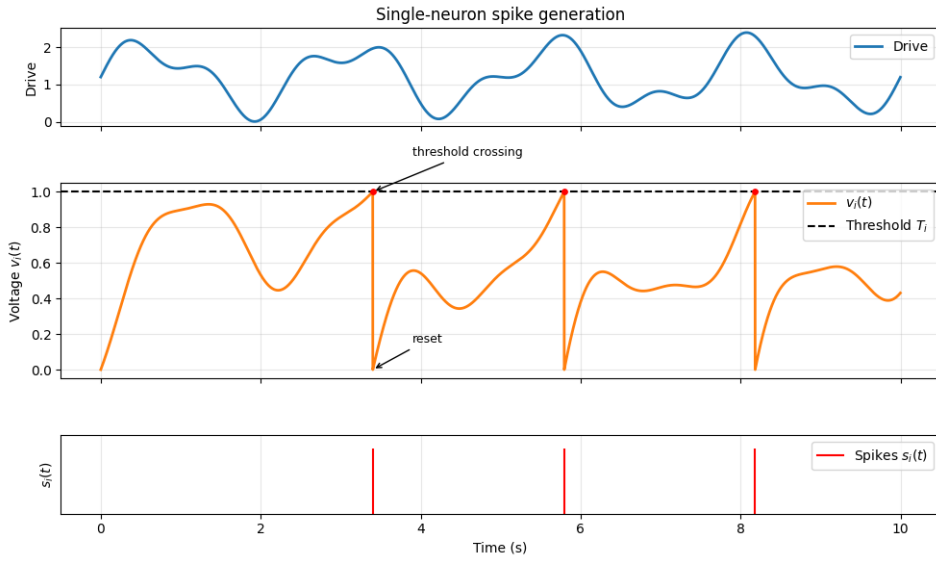
its neural representation. Individual spikes are first transformed into continuous synaptic traces and are subsequently decoded to reconstruct the represented state.

2.2.1 Neuron model

Each neuron in the population is modeled as a Leaky Integrate-and-Fire (LIF) neuron. Each neuron integrates incoming synaptic currents into its membrane potential, which evolves according to

$$\tau_m \frac{dV_i(t)}{dt} = -(V_i(t) - V_{\text{rest}}) + R_m I_i(t), \quad (5)$$

The membrane potential v_i integrates incoming inputs over time and emits a spike whenever it exceeds a threshold T_i . Following a spike, the membrane potential is reset.



The spike train of neuron i is represented by

$$s_i(t) = \sum_k \delta(t - t_i^k), \quad (6)$$

where t_i^k denotes the time of the k -th spike and $\delta(\cdot)$ is the Dirac delta function.

The collective spike activity of the population forms the basis for the network's representation of the dynamical system.

2.2.2 Voltage dynamics and spike threshold

Following Boerlin et al., spike generation is determined by an error-based membrane potential. In this framework, the voltage dynamics of neuron i are given by

$$\dot{v}_i(t) = -\lambda_v v_i(t) + D_i^\top (\lambda_v x(t) - \dot{x}(t)) - D_i^\top D s(t), \quad (7)$$

where λ_v denotes the membrane leak constant, D_i is the decoding vector of neuron i , $x(t)$ is the target state of the dynamical system, D is the full decoding matrix, and $s(t)$ is the vector of spike trains across the population. The first term describes the leak of the membrane potential, the second term provides feedforward drive from the target dynamics, and the final term represents the fast corrective interaction induced by

population spiking. This fast component ensures that emitted spikes rapidly reduce the instantaneous reconstruction error.

A neuron emits a spike whenever its membrane potential exceeds threshold,

$$v_i(t) > T_i, \quad (8)$$

with

$$T_i = \frac{1}{2} D_i^\top D_i. \quad (9)$$

After a spike, the neuron contributes to the population spike train $s(t)$ and thereby influences the subsequent voltage dynamics and decoded estimate. This voltage-based spike rule is used throughout the simulations; we refer to Boerlin et al. for the full derivation.

2.2.3 Synaptic filtering

Because spikes are instantaneous events, they must be transformed into continuous signals before they can be used to represent a dynamical state. This is achieved through synaptic filtering.

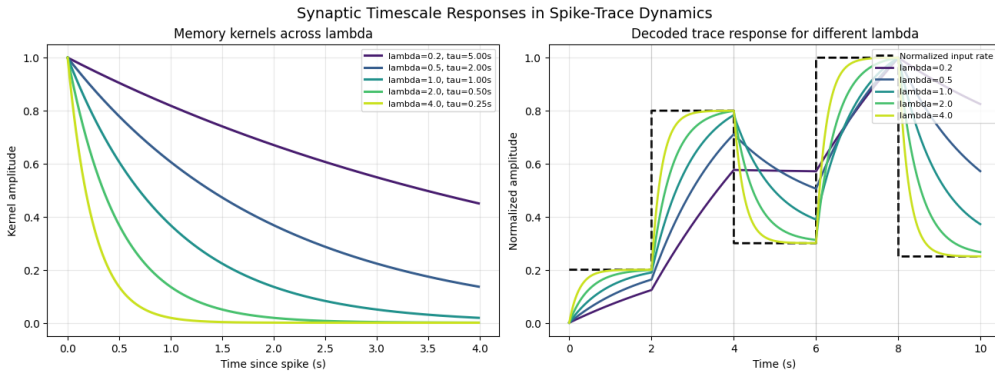
The filtered spike trains evolve according to

$$\dot{r} = -\lambda r + s, \quad (10)$$

where s denotes the vector of spike trains and λ controls the rate at which synaptic traces decay.

The filtered activity r provides a smoothed representation of recent spike history. Consequently, past spikes continue to influence the represented state after they occur, introducing memory into the network.

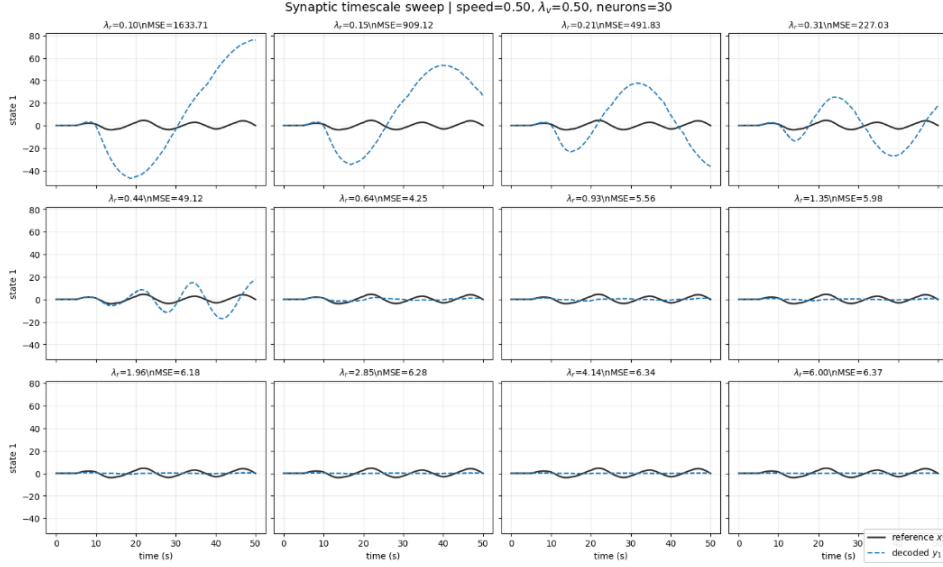
Equation (7) can be interpreted as a low-pass filter applied to the spike train. Rather than affecting the represented state instantaneously, each spike contributes an effect that gradually decays over time. The resulting synaptic trace therefore reflects a weighted history of recent neural activity, providing a continuous signal that can be decoded into an estimate of the system state.



2.2.4 The role of λ

The parameter λ determines the timescale over which past spikes influence the current synaptic state. Large values of λ cause traces to decay rapidly, resulting in a representation that closely follows incoming spikes and responds quickly to changing inputs. Smaller values of λ produce longer-lasting traces, increasing temporal integration and smoothing the decoded signal. However, this additional memory comes at the cost of increased delay,

which may reduce tracking accuracy and destabilize closed-loop dynamics. Understanding this trade-off forms the central focus of the experiments in this thesis.



2.2.5 Linear decoding

The filtered spike trains are converted into a continuous estimate of the represented state through a linear decoder:

$$y = Dr, \quad (11)$$

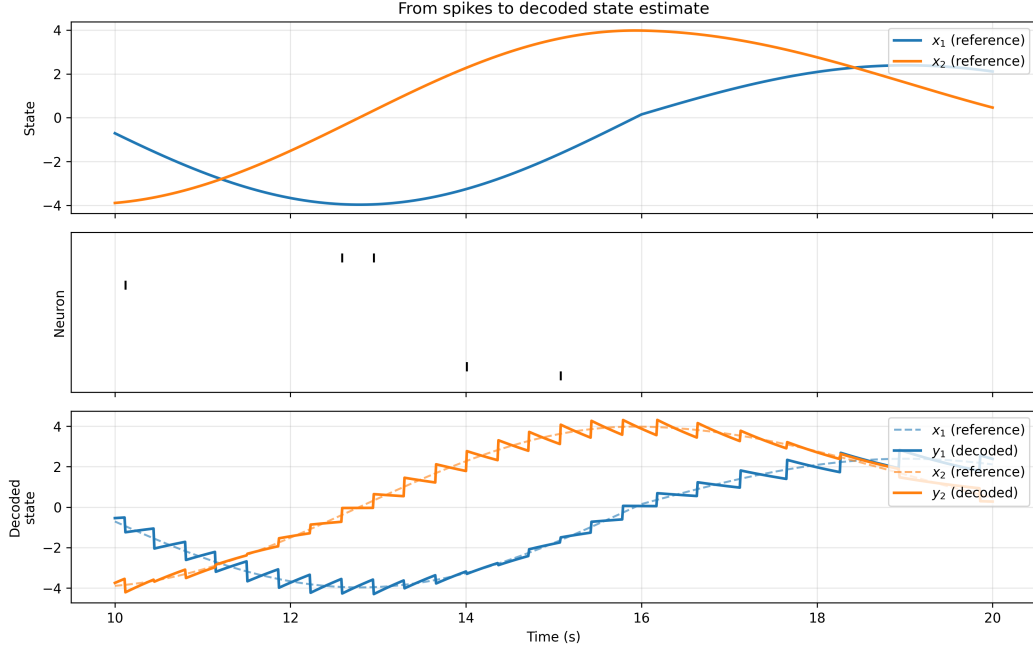
where $D \in \mathbb{R}^{2 \times N}$ is the decoding matrix and r contains the filtered spike trains.

Each column of D corresponds to the decoding vector of a single neuron. The decoding vectors are sampled randomly from a standard normal distribution and subsequently normalized to unit length. As a result, neurons are associated with different preferred directions in the two-dimensional state space while contributing equally in magnitude to the decoded signal.

The state estimate obtained through Eq. 11 approximates the true dynamical state:

$$y \approx x. \quad (12)$$

This distributed population code allows the network to represent continuous-valued dynamical states through the collective activity of many spiking neurons. Furthermore, the decoder remains fixed throughout all experiments, allowing changes in performance to be attributed primarily to the effects of synaptic filtering rather than modifications of the readout mechanism.



2.3 Simulation pipeline

The model is simulated using a discrete-time Euler integration scheme. At each timestep:

- update of Eq. (1),
- update of membrane potentials,
- spike generation and reset,
- update of Eq. (??),
- computation of feedback in the closed-loop setting.

In the closed-loop setting, $\mathbf{y}$ is fed back into Eq. (1), creating a coupled system.

2.4 Experimental manipulations

The main experimental manipulation is the synaptic time constant λ , which controls Eq. (??). We compare fast and slow regimes to study tracking performance and stability.

We also compare open-loop and closed-loop configurations (see Eq. (1)).

2.5 Evaluation metrics

Performance is evaluated using:

- tracking error between $\mathbf{x}$ and $\mathbf{y}$,
- stability of the dynamics,
- responsiveness and delay,
- spike statistics (rate, variability, sparsity).

3 Results

3.1 Baseline Dynamics and Tracking

Results of $Ax + Bu$ in simplest form to establish baseline. What happens if we change the leak. What does the spike plot look like. Run it through SCN, what does the decoded output look like and how does it perform. Establish $x \rightarrow Dr \rightarrow y$.

 `analog_trace.png`

3.2 Closed-Loop Feedback Dynamics

Now replace x with y (E.g. what happens when we go closed-loop). What happens to the dynamics when we change leak, what happens to spike statistics, etc. What does it do well and what not?

- Error
- leak
- Robustness
- trace trajectory
- stability
- spike statistics

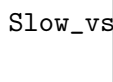
 `figures/Closed_loop.png`

Conclusion of what closed-loop feedback does to the dynamics of the system.

4 Slow filtering

Now apply the slow filtering r_s . *Explain mathematically how this works and why we do this. What is the effect on*

- Slow filter
- Synaptic Kernel
- Time constant
- low-pass filtering
- r and r_s *mathematically*

 `Slow_vs_Biological_synapse.png`

What is the trade-off between different metrics like delay/smoothness, robustness/lag, temporal integration vs stability.

4.1 Modified spike rule

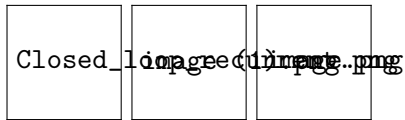
Explain why we need this and how (math) we do this, what does it improve, what does it do worse.

Mathematical explanation of:

- Treshhold modification
- Error prediction
- voltage equation

Evaluation on the metrics.

4.2 Figures and Tables



5 Discussion

Acknowledgements

References

Here you place the sources you used in your thesis. Please make sure you choose one specific style and stick to it consistently, both within the text and here in the bibliography. In most cases for AI theses that is either APA or IEEE style.

Assessment criteria: Cites and discusses relevant literature; uses an accurate and consistent referencing style; the reference list is complete.

Appendix